

Cost-aware deep allocation across published asset-pricing anomalies

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Abstract

We ask what is gained by training a neural network on the objective a fund is actually paid for — the Sharpe ratio of a book net of trading costs — rather than on squared forecast error. The asset universe is the 212 published cross-sectional anomalies of the Open Source Asset Pricing project, assembled point-in-time by publication date; the problem is how much capital to allocate to each, monthly. One permutation-equivariant architecture is trained under four objectives with data, folds and portfolio construction held fixed. Over 203 out-of-sample months (2008–2024) the ranking of models by gross Sharpe ratio and by net Sharpe ratio differs sharply: all nine models look comparable gross (0.94–1.38), but net of measured anomaly-specific costs only the two cost-aware objectives clear the equal-weighted benchmark, and five lose money outright. The model with the *highest* information coefficient earns a net Sharpe ratio of 0.19; the model with the *lowest* earns 0.45. No model beats $1/N$ significantly, but the cost-blind ones lose to it significantly. We show the mechanism is not the expected one — the cost-aware model does not avoid expensive anomalies, it stops rebalancing.

1 Introduction and problem statement

A machine-learning pipeline for asset returns usually minimises a pointwise loss and then, separately, sorts the resulting forecasts into a portfolio. The two steps optimise different things. Squared error weights an observation by its return variance rather than by the capital that can be deployed against it; it is indifferent to the cross-sectional ordering, which is what a portfolio actually monetises; and it knows nothing about turnover, so two models with identical forecast accuracy — one producing stable rankings, one reshuffling every month — are treated as equally good and are worth very different amounts of money.

Portfolio construction, however, is differentiable. Mapping scores to weights is a demeaning and a normalisation; turnover is an L_1 distance to last month’s drifted book. The entire path from features to net-of-cost profit can therefore be back-propagated through, and the network can be asked to maximise the quantity the fund is paid for. This paper measures what that is worth.

We deliberately study the problem one level above stock selection. The assets are the published anomalies themselves, so the allocation decision is the one a multi-strategy fund faces: given a menu of documented, replicated strategies, each with its own return, risk and *cost of being held*, how should capital be spread across them? This setting has a property that makes the cost question sharp. The intrinsic turnover of an anomaly’s underlying stock book varies by a factor of 500 — an annual-accounting signal barely moves, an analyst-revision signal is rewritten every month — so the cost of *holding* a sleeve differs enormously across the menu, quite apart from the cost of changing one’s mind about it.

Three hypotheses are tested, each isolating one change: whether seeing the whole cross-section helps (§3), whether optimising the ordering beats optimising the level, and whether optimising net-of-cost profit beats optimising the ordering.

Table 1: Measured monthly turnover of the underlying stock book, and the implied cost of carrying the sleeve for a year at a 25 bp one-way stock trading cost.

	Monthly turnover	Carrying cost
CitationsRD, GP, PatentsRD (annual accounting)	0.008	0.02 %/yr
Median anomaly	0.129	0.39 %/yr
dCPVolSpread, ChangeInRecommendation (options, analyst revisions)	3.98	12 %/yr

2 Data and feature engineering

Returns. The long-short monthly return of each of 212 predictors comes from `PredictorPortsFull` in the Open Source Asset Pricing release (Chen and Zimmermann, 2022). These are the assets we allocate across. Returns are reported in percent and are converted to decimals; the portfolios are equal-weighted decile spreads, self-financing, so no risk-free adjustment applies.

A point-in-time universe. An anomaly enters the investable universe in the January *after* the year its paper appeared. An investor in 1980 could not allocate to an anomaly documented in 2005, and a backtest that permits it is measuring a time machine rather than a strategy. The screen is expensive: it cuts 173,302 anomaly-months to 50,370 and moves the sample start to 1994, because fewer than 20 anomalies had been published before then. The universe grows from 22 assets in 1994 to 208 by 2020, which is itself an honest picture of what an allocator’s menu looked like through time. This construction also makes “years since publication” a legitimate feature rather than a leak, since by construction it is never negative.

Features. Twenty-eight continuous features are built from information dated t or earlier: trailing cumulative returns over 1, 3, 6, 12, 24, 36 and 60 months (including the 12-month-skipping-one construction), realised volatility and trailing Sharpe ratios, 60-month drawdown, skewness, and the anomaly’s beta and correlation to the equal-weighted index of all currently available anomalies. Twenty-one indicators capture the strategy’s own definition — economic category, data source, discrete versus continuous form, value versus equal weighting — and nineteen missingness flags mark where a feature could not be computed. Continuous features are ranked within each month and rescaled to $[-1, 1]$; indicators are left alone, since ranking a binary column produces a number whose value depends on how many peers share the category. Every transform is cross-sectional *within a month*, so no test-period distribution can leak into a training-period feature.

Variables that exist in the metadata but are not knowable at time t are excluded, even though they predict returns well: Chen and Zimmermann’s replication-quality grades and the 2025 citation counts are assessments made in the 2020s. Raw publication years are also dropped — with only 212 assets they come close to naming each anomaly, and a network given them can memorise which specific anomalies paid off instead of learning what *kind* of anomaly pays.

Costs, measured rather than assumed. Allocating across anomalies incurs two distinct costs. *Switching* is paid on the change in allocation weight. *Carrying* is paid on the weight held, because the sleeve rebalances its own stock book every month whether or not the allocator touches it. The second is not published, but it is recoverable: the release also distributes the firm-level signal behind each portfolio, so we rebuild each anomaly’s decile book month by month and measure how much of it turns over. Table 1 gives the range.

The estimate is used causally: at month t the model and the cost model see an expanding-window average of turnover realised strictly before t . This costs nothing in accuracy, because turnover is close to a structural constant of each signal — its cross-decade correlation is between 0.985 and 0.997, since it is determined by the sampling frequency of the underlying data. Anomalies

documented with multi-month holding periods roll only part of the book each month, so their measured churn is divided by the holding period.

3 Models

Benchmarks. The classical ladder — Huber-loss OLS, ridge, elastic net, partial least squares and gradient-boosted trees — is fitted pooled, with each hyper-parameter chosen on the fold’s own validation block. The **equal-weighted** ($1/N$) **allocation** is treated as a model and appears as a row in every table, costed identically. It is famously hard to beat (DeMiguel et al., 2009), and burying it in the text is how that fact gets hidden.

Networks. **AssetMLP** is the Gu et al. (2020) benchmark: a feed-forward network applied to each asset-month independently, which cannot see the rest of the cross-section. **CrossSectionalNet** is a Set Transformer (Lee et al., 2019): assets exchange information through 32 learned inducing points, costing $O(Nm)$ rather than the $O(N^2)$ of full self-attention. It is permutation-equivariant by construction — relabelling the assets permutes the outputs identically — so it can only learn from an asset’s characteristics and its peers’ distribution, never from its position in the input. This is verified numerically in the test suite to 3.6×10^{-7} . At 19,457 parameters it is deliberately small; the panel has 48,236 rows.

Allocation. Scores become weights as $1/N$ plus a dollar-neutral tilt of fixed L_1 size. The benchmark an allocator must beat is equal weighting, so the network is asked for the deviation from it; this also conditions training well, since at initialisation the scores are noise and the portfolio starts life as the benchmark rather than as a random book. A per-anomaly cap of 10% is enforced by fixing capped names and redistributing only across free ones, and the book is scaled to a 10% ex-ante volatility target using a strictly lagged trailing estimate.

Objectives. With weights w_t set at the close of month t and earning r_{t+1} ,

$$\tilde{w}_{t-1} = \frac{w_{t-1} \odot (1 + r_t)}{1 + w_{t-1}^\top r_t}, \quad (\text{drifted book}) \quad (1)$$

$$c_t = \sum_i s_i |w_{t,i} - \tilde{w}_{t-1,i}| + \sum_i k_i |w_{t,i}|, \quad (\text{switching} + \text{carrying}) \quad (2)$$

$$\mathcal{L}_{\text{Sharpe}} = -\sqrt{12} \frac{\overline{(w^\top r - c)}}{\text{sd}(w^\top r - c)}. \quad (3)$$

Charging against the *drifted* book matters: an allocation left untouched still changes as sleeves earn different returns, and billing that as a trade invents turnover that never happened. The mean–variance variant replaces the Sharpe ratio with $\bar{x} - \frac{\gamma}{2}\text{var}(x)$, which has the same optimum at a given risk level but a far better-conditioned gradient when the window is short. Two cost-blind objectives — masked squared error and the cross-sectional rank correlation — are trained on the identical architecture, and each model is early-stopped on *its own* validation objective, so no model borrows another’s advantage.

4 Evaluation methodology

The sample is walked forward one year at a time. Each refit trains on everything available up to a four-year validation block used for early stopping and hyper-parameter selection, then predicts the following year, which it has never seen. A **12-month embargo** separates training from validation, because several anomalies are built from annual accounting data and a shorter gap would place the

Table 2: Out-of-sample performance, 2008-02 to 2024-12 (203 months), net of measured anomaly-specific costs. IC is the mean monthly rank information coefficient. Break-even is the one-way stock trading cost at which the net return reaches zero.

Model	Objective	IC	Gross SR	Net SR	t	Turnover	Break-even
<code>xs_utility</code>	net-of-cost utility	0.063	1.38	0.45	1.53	0.67	183 bp
<code>xs_sharpe</code>	net-of-cost Sharpe	0.066	1.21	0.40	1.48	0.62	179 bp
<code>equal_weight</code>	$1/N$ benchmark	—	1.32	0.39	1.44	0.09	866 bp
<code>xs_ic</code>	rank correlation	0.101	1.17	0.19	0.82	0.91	115 bp
<code>nn3_mse</code>	squared error, per-asset	0.087	1.07	0.17	0.73	0.88	113 bp
<code>gbrt</code>	squared error, trees	0.095	1.03	−0.01	−0.03	1.17	81 bp
<code>xs_mse</code>	squared error	0.092	1.07	−0.02	−0.08	1.17	85 bp
<code>pls</code>	squared error, PLS	0.087	0.96	−0.03	−0.15	1.05	83 bp
<code>ols_h</code>	Huber, linear	0.093	1.04	−0.07	−0.29	1.23	77 bp
<code>ridge</code>	squared error, linear	0.086	0.97	−0.10	−0.42	1.17	75 bp
<code>enet</code>	squared error, sparse	0.080	0.94	−0.30	−1.21	1.54	55 bp

same fundamentals on both sides of the boundary (López de Prado, 2018). Seventeen folds produce 38,539 out-of-sample predictions per model, covering 2008-02 to 2024-12 without overlap; the test suite asserts these properties mechanically rather than trusting the implementation. Neural results are averaged over five seeds, since a single network fitted to monthly returns is a high-variance estimator.

We report statistical and economic metrics separately because they can disagree, and here they do. Out-of-sample R^2 follows Gu et al. (2020) in benchmarking against a zero forecast rather than the historical mean. Sharpe ratios carry Lo (2002) standard errors and Newey–West t -statistics. The *break-even cost* — the one-way cost at which a strategy stops making money — is reported because it is assumption-free.

5 Results

Gross and net rank the models differently. Table 2 is the paper’s main result. Gross Sharpe ratios span 0.94–1.38 and every model is broadly comparable; a study that stopped there would report nine successes. Net of costs, only the two cost-aware objectives clear the benchmark and five models lose money outright. The spread between best and worst net Sharpe ratio is 0.75, of which almost none is visible gross.

Predictive accuracy does not order the models by economic value. `xs_ic` attains the highest information coefficient (0.101) and the highest out-of-sample R^2 (1.48%) of any model and earns a net Sharpe ratio of 0.19. `xs_utility` has the *lowest* IC (0.063) and the best net Sharpe ratio. The two extremes are therefore inverted — but we resist the stronger claim: across all ten models the Spearman correlation between IC and net Sharpe ratio is only -0.14 ($p = 0.70$, $n = 10$), which is indistinguishable from zero. The defensible statement is the weaker and still useful one: **knowing which model forecasts best tells you nothing about which model earns most.** Improving forecast accuracy and improving the traded objective are different problems, and a model-selection procedure that ranks on IC is choosing close to at random.

The honest significance statement. Regressed on the benchmark, `xs_utility` has an alpha of +1.46%/yr with a HAC t of 0.88 and a beta of 1.21, so part of the raw gap is leverage rather than skill; its information ratio is 0.31 and the t on its active return is 1.15. **No model beats $1/N$ at conventional significance.** What *is* significant is the damage done by the cost-blind objectives: `ridge` ($t = -1.97$), `xs_mse` (-2.08) and `enet` (-2.55) destroy value against equal

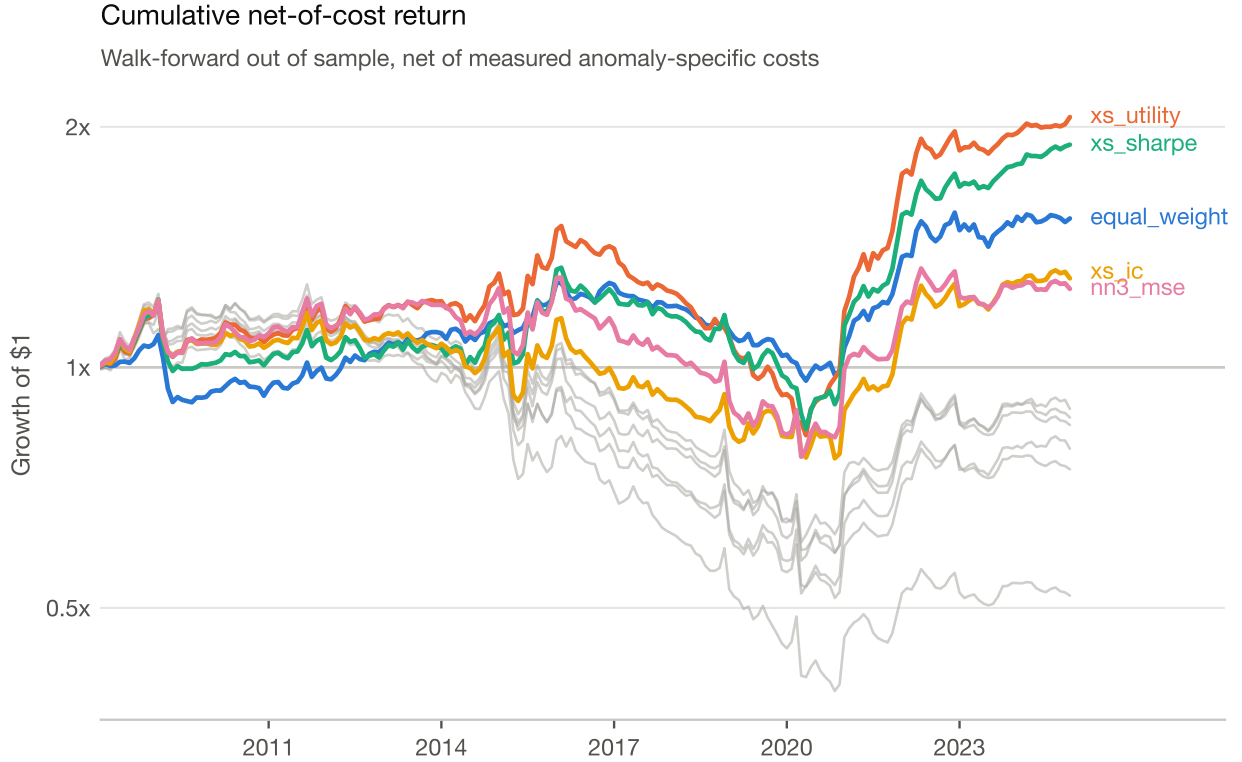


Figure 1: Cumulative net-of-cost return, walk-forward out of sample. Grey lines are the seven models not named.

weighting. The defensible claim is therefore not that machine learning beats naive diversification; it is that the choice of objective decides whether ML allocation loses to $1/N$ or merely ties it.

The mechanism is not the expected one. The natural hypothesis is that a cost-aware model learns to avoid anomalies that are expensive to carry. It does not. Sorting anomalies into quintiles by measured carrying cost, the best model’s average tilt away from $1/N$ is -0.0007 on the cheapest quintile and $+0.0004$ on the dearest — flat, and if anything mildly the wrong way. What it learns is *allocation persistence*. Turnover falls from 0.88–1.54 to 0.62–0.67 and the annual switching bill from 5.3–9.2% to 3.7–4.0%, while carrying cost is similar for every model (3.9–6.1%) because all of them end up holding close to the full menu. Figure 2 shows the decomposition: the edge is in not rebalancing, not in what is held.

Robustness. Figure 3 scales both cost components together. At zero cost the models and the benchmark are nearly tied (1.32–1.38), confirming that essentially all of the observed differentiation is a cost phenomenon. The ordering in Table 2 is stable across the whole grid, and the cost-aware objectives degrade most slowly. Splitting the sample at 2015, **xs_utility** earns net Sharpe ratios of 0.47 and 0.45 in the two halves while **enet** deteriorates from 0.00 to -0.44 ; the cost-aware result is not driven by one sub-period.

6 Capacity: does any of this survive in tradable stocks?

Published anomaly returns are concentrated in small, illiquid names, so a result obtained on the unrestricted universe may describe a portfolio no fund could run. The release distributes every anomaly rebuilt using only stocks above the NYSE 20th percentile of market capitalisation; we repeat the entire study on that universe, refitting every model on its own walk-forward folds.

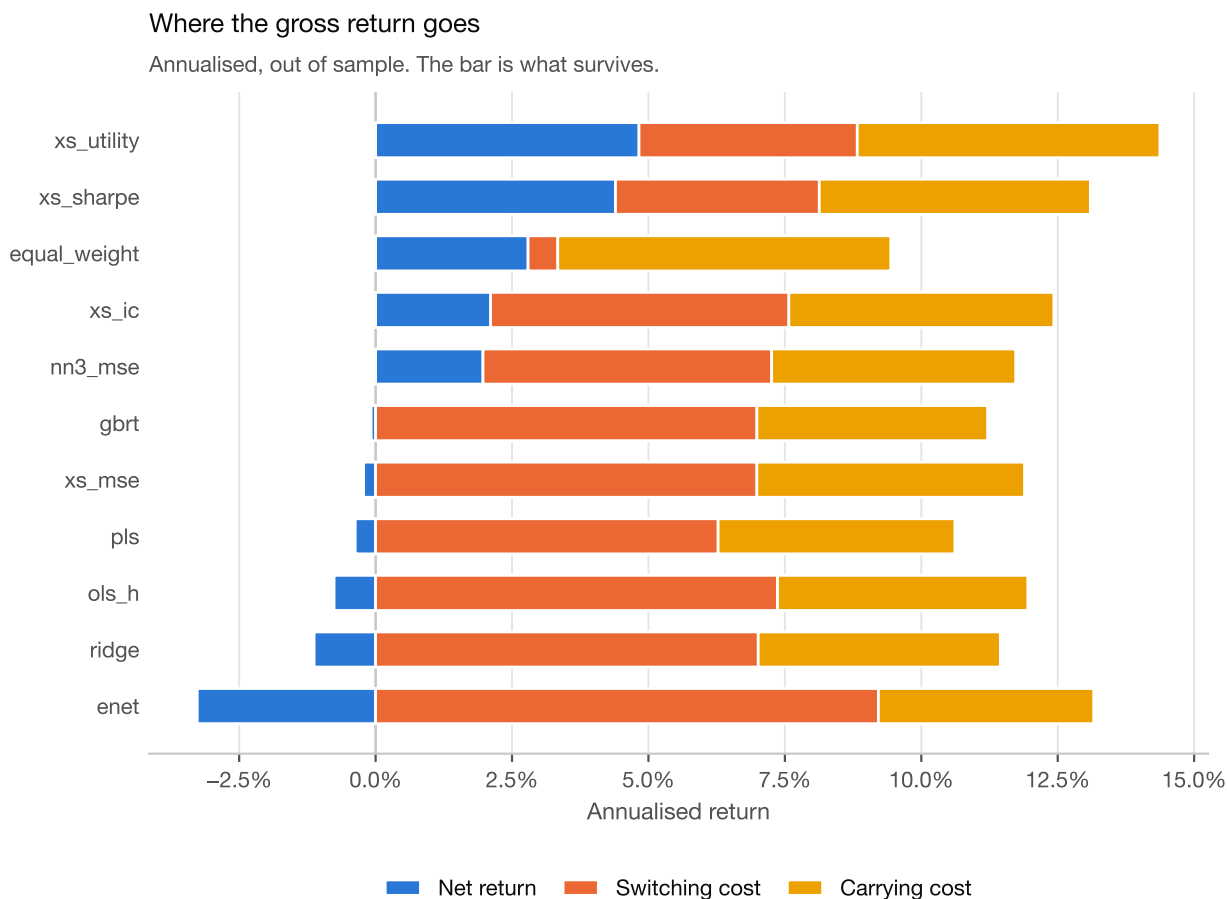


Figure 2: Annualised gross return split into switching cost, carrying cost and what survives.

The restriction is expensive before a single basis point of cost. The mean long-short return falls from 0.499% to 0.348% per month, and at zero assumed cost the benchmark's Sharpe ratio falls from 1.32 to 0.80 — a 40% haircut. Table 3 shows what is left after costs.

Two things follow, and the second is the more important.

The ranking of objectives survives; the edge does not. The rank correlation of net Sharpe ratios across the two universes is +0.75: cost-aware objectives still beat cost-blind ones, and **enet** is still last. The paper's claim about *objectives* is therefore not an artefact of an untradable sample. But every point in Figure 4 lies below the 45-degree line and every one is negative — at 25 bp, nothing in the tradable universe makes money, benchmark included.

Where the model's advantage does survive, it reverses. At 12.5 bp, a defensible cost for large capitalisations, the best allocation in the tradable universe is **equal weighting** (0.34), ahead of **xs_utility** (0.26) and **xs_sharpe** (0.20). The cost-aware models beat 1/N in the universe where anomalies are strong and expensive, and lose to it in the universe where they are weak and cheap. Read together with §5, the honest summary of this project is narrow: *the choice of training objective reliably separates good ML allocation from bad, and no version of it reliably beats doing nothing.*

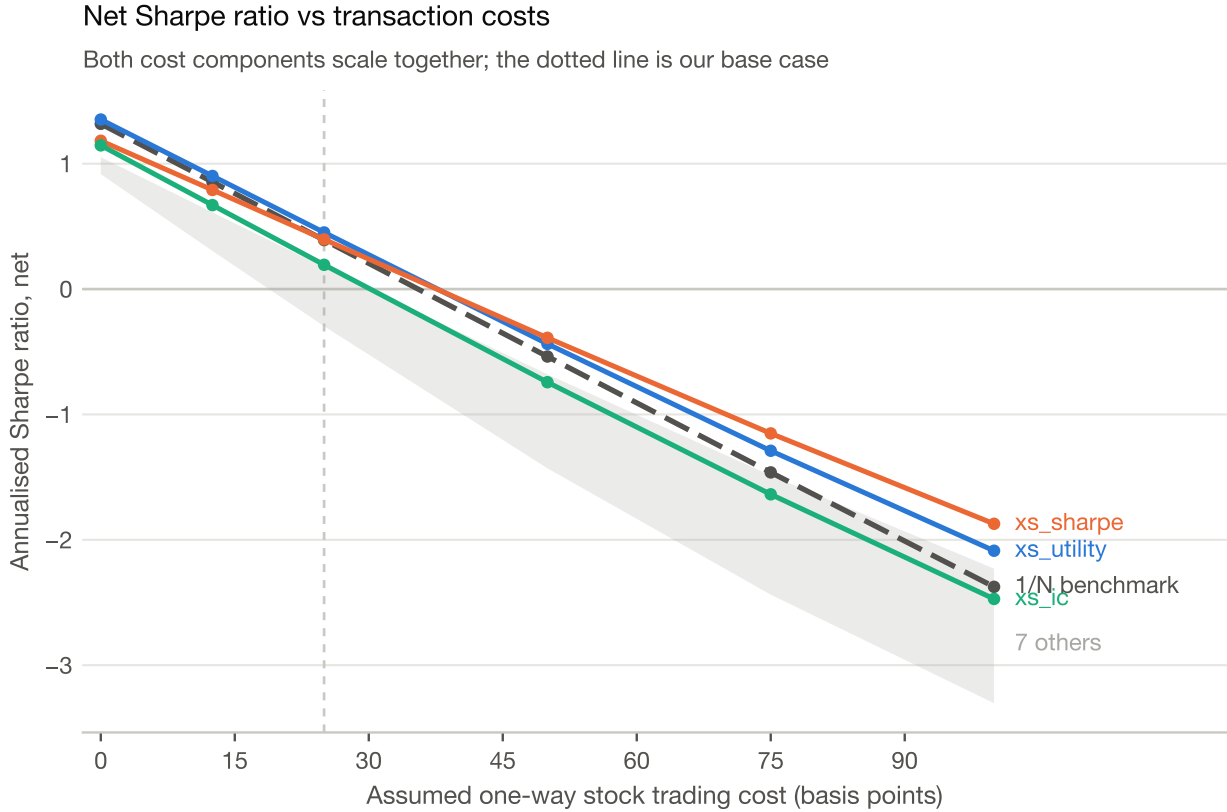


Figure 3: Net Sharpe ratio as both cost components are scaled. The dotted line is the 25 bp base case; the grey band spans the seven unnamed models.

7 Conclusion and extensions

Training a network on net-of-cost portfolio profit rather than on forecast error changes an allocation that loses to equal weighting into one that ties it, and it does so through a mechanism — refusing to rebalance — that a pointwise loss cannot express. The negative half of this finding is the statistically solid half: cost-blind machine learning applied to anomaly allocation destroys value against a benchmark that requires no model at all.

Three extensions follow directly. First, the no-trade band: our allocator rebalances every month by construction, whereas the natural response to a turnover penalty is a hold region, and solving for it explicitly should widen the gap this paper measures. Second, a lower-cost tradable universe: §6 shows the large-cap result hinges on the assumed spread, and a proper treatment would estimate the spread per anomaly from the actual market capitalisations it holds rather than applying one number to the whole universe. Third, sample length: 203 out-of-sample months cannot resolve a 0.06 difference in Sharpe ratio, and the point-in-time publication screen is what costs us the earlier data — relaxing it would be dishonest, but reporting the hindsight-universe result alongside would bound how much the screen costs.

Code and reproducibility. All data is free and public; `make all` reproduces every number from a clean clone. The codebase is 20 modules with 52 tests, built around synthetic panels where the answer is known: the pipeline must recover a planted signal and must find nothing in pure noise. Libraries used: PyTorch, scikit-learn, LightGBM, Polars, pandas, NumPy, SciPy, statsmodels, Matplotlib.

Table 3: Net Sharpe ratio in both universes. The right-hand columns sweep the assumed one-way stock trading cost; 12.5 bp is arguably the right figure for a large-capitalisation universe, and 25 bp the right one for the unrestricted sample.

Model	All stocks		Large caps only	
	12.5 bp	25 bp	12.5 bp	25 bp
xs_utility	0.90	0.45	0.26	-0.25
xs_sharpe	0.79	0.40	0.20	-0.28
equal_weight	0.86	0.39	0.34	-0.12
xs_ic	0.67	0.19	0.04	-0.59
enet	0.30	-0.30	-0.13	-0.85

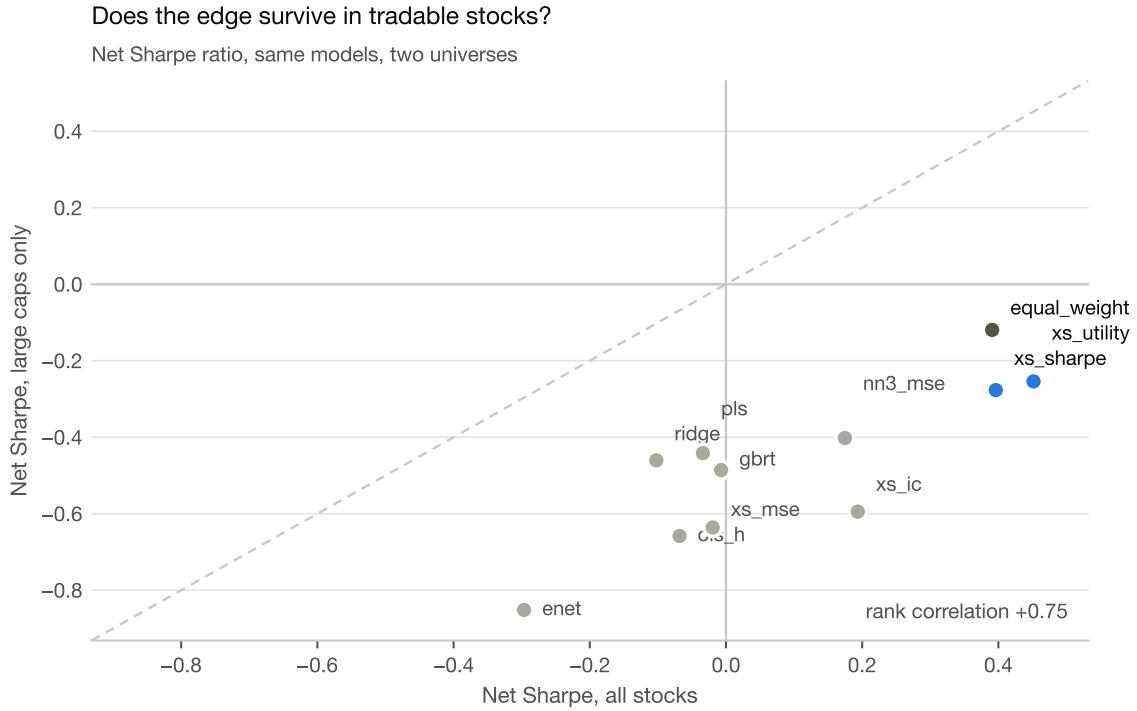


Figure 4: Net Sharpe ratio of the same models in both universes, at the 25 bp base case. Every model falls below the 45-degree line.

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